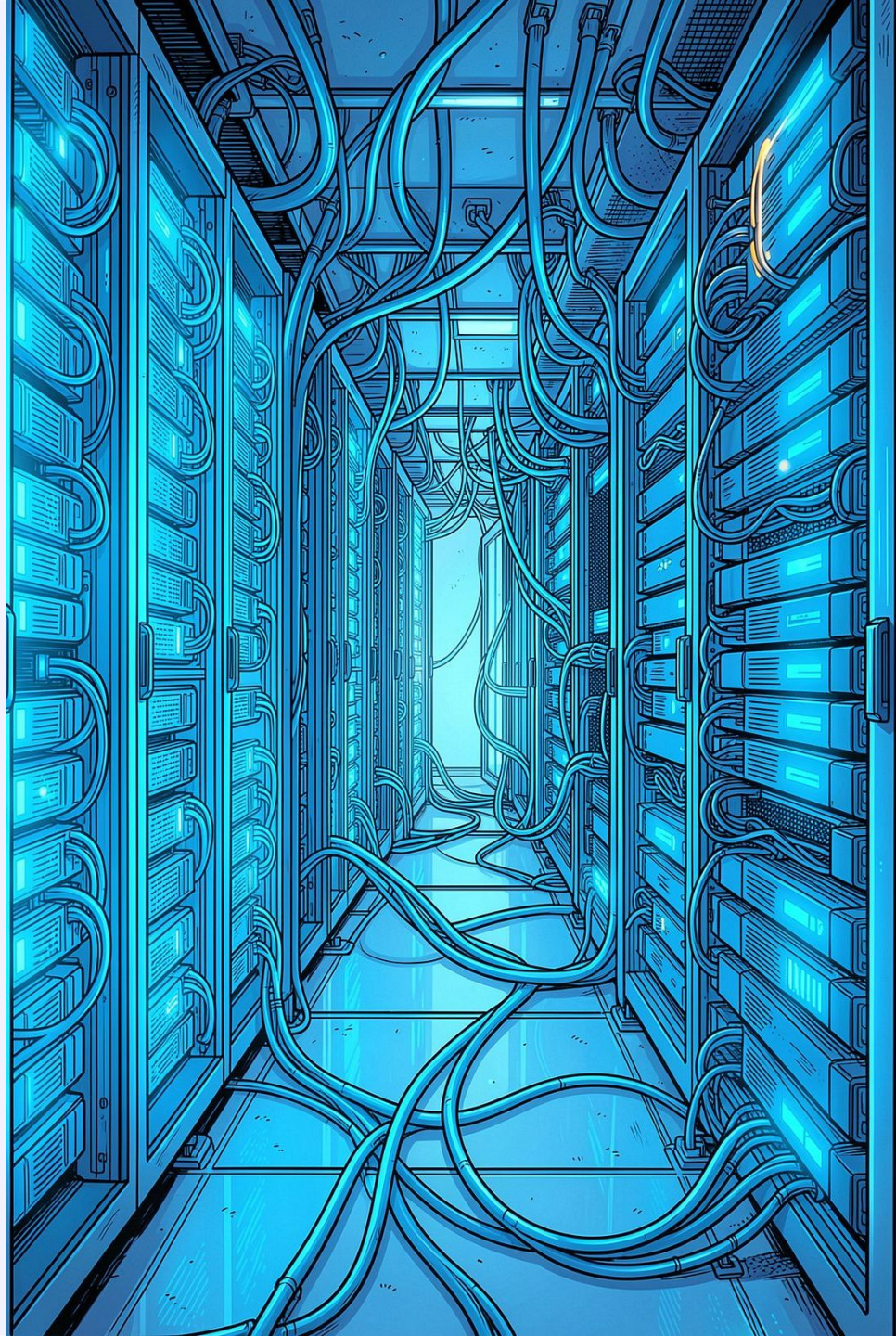
Azure Monitor and Log Analytics Workspace Design

Centralized Monitoring and Failure Detection for Azure Resources

TEAM MEMBER NAME:

- S.VATHSAN-2400031044
- SAHITH - 2400031082
- CH SASI DHARA GANESH-2400031134
- P.YASWANTH-2400031150



The Monitoring Challenge in Azure

Azure environments generate logs and metrics across many services, making it hard to track issues in one place.

Fragmented Visibility

Hard to correlate events across resources.

Misleading Metrics

Healthy CPU can still hide failures.

Project Objectives



Centralize Logs

Create a Log Analytics Workspace for unified data collection.



Analyze with KQL

Use Kusto Query Language to detect patterns and issues.



Automate Alerts

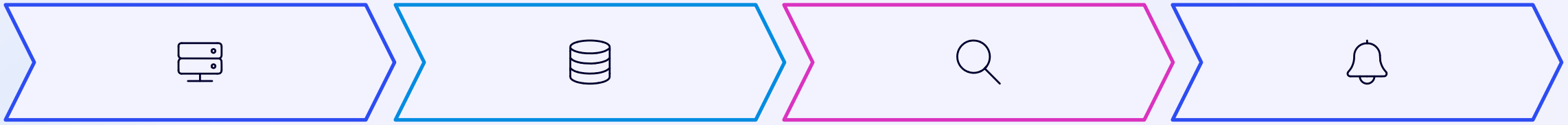
Set up Azure Monitor alerts and Action Groups.



Unified Dashboard

Build a centralized monitoring view.

End-to-End Monitoring Workflow



Azure Resources

Log Analytics Workspace

KQL Analysis

Azure Monitor Alerts

System Architecture

Architecture Layers

01

Azure Resources

VMs, App Services, Databases, Storage

02

Diagnostic Settings

Route logs and metrics to workspace

03

Log Analytics Workspace

Centralized data repository

04

Alerts & Notifications

KQL detection, monitor rules, action groups, email

Technologies & Implementation

Core Technologies

- Azure Monitor
- Log Analytics
- KQL
- Alerts & Action Groups

Implementation Steps

01 Resource Group

02 Workspace

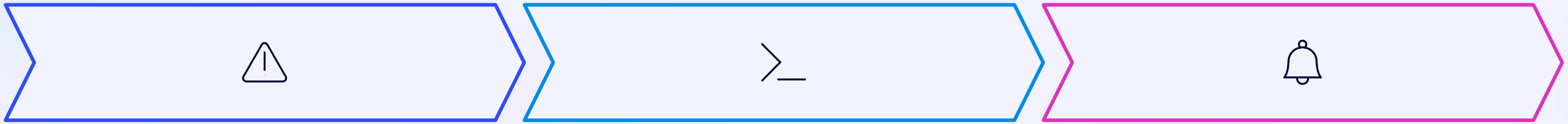
03 Connect Resources

04 Alerts

KQL Failure Detection

```
AzureActivity | where ActivityStatusValue == "Failed"
```

Detection Pipeline



Failure Occurs

KQL Query Runs

Alert Triggered

Cost & Data Residency

💰 Cost Optimization

- Collect Only Required Logs
- Monitor Ingestion Volume

🌍 Data Residency

- Select Workspace Region
- Consider Compliance Requirements

Testing & Results

Test Scenarios

✅ Success

No alert triggered — system operating normally

❌ Failure

KQL detects issue → Alert fires → Notification sent

Key Results

- Centralized Monitoring
- Faster Detection
- Automated Alerts

Conclusion & Future Scope

Centralized Azure monitoring enables **log analysis, failure detection, and automated alerting** across ~40 resources — reducing mean time to detect and improving operational resilience.

Future Scope

- AI anomaly detection
- Microsoft Sentinel
- Automated remediation
- Predictive monitoring
- Power BI integration

> **Thank You** — Questions & Discussion Welcome